

heatmap-forge

Generates a full GitHub activity heatmap on an existing remote repository by creating backdated commits that quietly toggle the `README.md` content. The README always ends up back to normal, so nothing stays corrupted.

How it works

- Reads all settings from a `.env` file in the project root (see `.env.example`).
- Uses the GitHub Contents + Git Data APIs to build a chain of commits directly on the remote repo — no local clone needed.
- Each backdated commit appends a temporary (invisible HTML-comment) block of words to the end of the README, then the **next** commit removes it. The file always returns to its original content by the final commit.
- If the repo has no `README.md`, an initial "add README.md" commit creates one, and the heatmap fills on top of it.
- Multiple `schedule` patterns control how many commits land on each day.

Project structure

```
heatmap-forge/  
├── index.js           # entry point / orchestrator  
├── .env               # your config (git-ignored, stays local)  
├── .env.example       # documented template for .env  
├── src/  
│   ├── config.js      # loads .env and validates required values  
│   ├── content.js      # README content generation + commit messages  
│   ├── github.js       # GitHub API client (Contents + Git Data endpoints)  
│   ├── scheduler.js    # per-day commit frequency logic  
│   └── utils.js        # random helpers + tiny .env loader  
└── package.json
```

Requirements

- Node.js (any modern version).
- A GitHub **Personal Access Token** with `repo` scope, or a fine-grained token with **Contents: Read & Write** on the target repo.
- The target repo must already exist on GitHub. `owner/repo` format.

Setup

1. Install/ensure Node.js.
2. Copy the example env and open it:

```
copy .env.example .env
```

3. Edit `.env` to fill in your values (see [Configuration](#)).

4. Run:

```
npm start
```

Configuration

Everything is configured via `.env` at the project root. The `.env` file is what the tool reads; a documented template lives in `.env.example`.

Key	Required	Description
<code>GITHUB_TOKEN</code>	yes	GitHub PAT (or fine-grained token). See token notes below.
<code>REPO</code>	yes	Target repo in <code>owner/repo</code> format. Must already exist.
<code>START_DATE</code>	yes	First commit date, <code>YYYY-MM-DD</code> . Can be in the past.
<code>END_DATE</code>	yes	Last commit date, <code>YYYY-MM-DD</code> . Must be <code>>= START_DATE</code> .
<code>BRANCH</code>	no	Branch to commit on. <code>main/master</code> auto-detect the default branch.
<code>COMMITTS_PER_DAY_MIN</code>	no	Minimum commits on an active day (default <code>1</code>). Must be <code>>= 1</code> .
<code>COMMITTS_PER_DAY_MAX</code>	no	Maximum commits on an active day (default <code>3</code>). Must be <code>>= min</code> .
<code>SCHEDULE</code>	no	Frequency pattern (default <code>dense</code>). See table below.
<code>AUTHOR_NAME</code>	yes	Name stamped on each commit.
<code>AUTHOR_EMAIL</code>	yes	Email stamped on each commit. Should match your GitHub account email so squares count toward your heatmap.

Numbers left empty fall back to sensible defaults; `BRANCH/SCHEDULE` default to `main/dense`.

`SCHEDULE` values

Value	Behavior
<code>dense</code> (default)	Commit on every day, <code>min..max</code> commits each. Solid heatmap.
<code>sparse</code>	Commit on only ~65% of days. Lighter, patchier heatmap.
<code>weekday</code>	Commit on weekdays only (no weekends). Mon–Fri heatmap.

Value	Behavior
<code>realistic</code>	Heavy weekdays, light weekends, occasional skipped days. Most natural-looking.

Token

In `.env`, set either `GITHUB_TOKEN` (recommended) or `TOKEN`. If left blank, the tool also checks the `GITHUB_TOKEN` environment variable, then the `--token=xxx` CLI argument.

Use a classic token with the `repo` scope, or a fine-grained token with

Contents: Read & Write on the target repo.

Example `.env`

```
GITHUB_TOKEN=ghp_xxxxxxxxxxxxx
REPO=octocat/hello-world
START_DATE=2024-01-01
END_DATE=2024-12-31
BRANCH=main
COMMITES_PER_DAY_MIN=1
COMMITES_PER_DAY_MAX=3
SCHEDULE=realistic
AUTHOR_NAME=Octo Cat
AUTHOR_EMAIL=octocat@example.com
```

Verification

After running, check the result:

- <https://github.com/<owner>/<repo>> — the contributions heatmap should be filled over your chosen date range.
- The commit history on GitHub should show author dates falling within range.

Notes & caveats

- Commits are **detected on GitHub** based on the *author date*, which this tool sets to random times within each day — that's what makes past squares light up.
- The tool **cannot** backfill activity for dates before your GitHub account existed, and GitHub may flag the unusual all-at-once pattern. Use sensible schedules (`sparse`, `realistic`) for a less suspicious result.
- Committing to repositories you don't own or that prohibit this may violate their usage terms. Use only on repos you control.
- Add `.env` to `.gitignore` so your token is never committed.